

Flutter Profile Card UI

Assignment Report by Gaurav Kulkarni

Part 1: What I Understood and Learned

For this assignment, the goal was to transition from writing logic in Dart to actually building a visual interface in Flutter. The objective was to design a clean profile card screen using core layout and UI widgets. It was really interesting to see how Flutter uses a widget tree to build everything visually, which felt very different from my usual backend work with Python and C++.

Layouts with Column, Row, and Container

I finally got a solid understanding of how Flutter structures layouts. I learned that the Column widget is used to stack elements vertically (like putting the profile picture above the user's name), while the Row widget aligns things horizontally (like placing the follower count next to the following count). I also used the Container widget extensively to add padding, margins, and rounded borders to the profile card itself so it didn't just stretch across the whole screen.

Styling with CircleAvatar, Text, and Icons

To make the card look like a real profile, I used the CircleAvatar widget, which made it super easy to display a circular profile picture without having to manually crop an image. I combined this with Text widgets for the name and bio, and learned how to use TextStyle to change font sizes and weights. Adding Icon widgets (like a little mail icon for the email address or a location pin) really helped the UI look polished and professional.

Applying Custom Theme Colors

Instead of sticking to the default Flutter blue, I learned how to apply custom theme colors. I figured out how to use hex codes with the Color(0xFF...) format and how to wrap my design in a custom theme. This allowed me to give the background, card, and text a cohesive color palette that matched the aesthetic I was going for.

Part 2: Problems Faced and How I Fixed Them

Problem 1: The dreaded RenderFlex Overflow error

When I first added the user's bio text into a Row next to an icon, the text was too long for the screen. Flutter threw a massive yellow and black striped error on the emulator saying 'A RenderFlex overflowed by X pixels'. **How I fixed it:** I did some research and learned about the Expanded widget. By wrapping my Text widget inside an Expanded widget, it forced the text to wrap to the next line instead of trying to push off the edge of the screen.

Problem 2: Everything was squished to the top left

When I first put my widgets into Columns and Rows, they all clumped together at the very top left of the screen, making the profile card look terrible and unorganized. **How I fixed it:** I discovered `mainAxisAlignment` and `crossAxisAlignment`. By setting my Row's main axis alignment to `MainAxisAlignment.spaceEvenly`, it perfectly spaced out the profile stats. I also used `crossAxisAlignment` to center the profile picture and text in the middle of the Column.

Conclusion

This assignment was a really rewarding introduction to frontend development with Flutter. It was incredibly satisfying to write code and instantly see a sleek UI pop up on the emulator. Figuring out how Rows and Columns interact, and learning how to fix layout overflows, gave me a lot of confidence for building more complex app screens in the future.